Jake K. Mann

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Work Experience

Dynamic Motion Control, Inc. — **San Diego, CA**
Automation Engineer — *August 2024-Present*

- Rewriting pathing program for ABB IRB 6600 6-DOF robot to reduce client's mold pour time by 37%.
- Developing sales proposals for robotics and automation, securing purchase orders of more than $180k.
- Programming Siemens PLCs and HMIs across projects in Robotics, Defense, and Biotech sectors.

Duke University Design Hub — **Durham, NC**
Project Manager — *February 2021-May 2024*

- Managed a team of undergraduate engineers to design and produce consumer product prototypes and research-grade lab devices, including automatic 'smart' syringes and testing rigs to quantify organic tissue strength.
- Taught advanced Fusion 360 CAD/CAM user courses and fabrication machine training.
- Provided Duke students and faculty with guidance and resources with 3D printing, laser/waterjet cutting, and CNC machining.
- Advised on manufacturability and performance during design sprints, primarily using Fusion360 and Solidworks.

Clark Construction Group — **Washington, D.C.**
Summer Associate — *Summers of 2022 and 2023*

- Coordinated Mechanical, Electrical, and Plumbing for renovations to Walter Reed Medical Center.
- Completed punchlist in paint, drywall, flooring, and MEP for 1,200 rooms.

Education

M.Eng. Mechanical Engineering, Class of 2024 — **Duke University**
Concentration in Autonomous, Intelligent Systems and Machines — *GPA: 4.0/4.0*

BSE Mechanical Engineering, Class of 2023 — **Duke University**
Focus in Robotics

Featured Projects- See ePortfolio

Rexy the T-Rex — *Spring-Fall 2023*

- Designed and 3D-printed bipedal T-Rex robot with onboard Raspberry Pi, servos and battery.
- Programmed sinusoidal locomotion patterns in Python to achieve basic bipedal movement.
- Optimized gait with Deep Q Reinforcement Learning via the OpenAI gym environment and PyTorch.

Getaway Mobile Smart Home - Central Monitoring System — *Fall 2022-Spring 2023*

- Developed an integrated smart-home control system using ESP32 sensor nodes and Raspberry Pi gateway architecture.
- Built a centralized monitoring and automation platform with HomeAssistantOS, aggregating power, lighting, temperature, and water-level telemetry into a unified control interface.

Compression of Neural Radiance Fields (NeRF) — *Fall 2024*

- Applied quantization and iterative node pruning to enable efficient NeRF inference on consumer-grade GPUs.
- Reduced model size by over 75% while preserving up to 90% visual fidelity, evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM).

Skills

Software: RobotStudio, SolidWorks, Fusion 360, ROS, Siemens TIA Portal, TwinCAT 3, WinCC Unified/Pro, Adobe Design Suite
Technical: CAD/CAM, Deep Learning for Robotics, Microcontrollers, FEA, CFD, IoT Sensors, State Machines
Manufacturing: 3D Printing, Laser Cutting, Waterjets, CNC (Tormach, ShopBot), Metal/Plastic Fabrication
Programming: Python, MATLAB, RAPID, Linux, C++, LaTeX
Languages: Spanish (Conversational / B2 Reading + Writing)